

Lab 6 Fall 2026

Linked List of Spotify Track Objects



CONCEPTS FOR LAB 6

- Linked Lists
- Classes/Objects
- Parsing .csv files

THIS IS NOT A PAIRED PROGRAMMING ASSIGNMENT

All students are to complete this assignment on their own. You all can talk to each other but you definitely shouldn't have the same code! If you do, that is considered cheating for this assignment (refer to syllabus academic integrity policy).

LAB 6 POLICIES

Generative AI Policy for Lab 6

You are **NOT** allowed to use GitHub Copilot or any other AI generation tool for **code generation**. Therefore, if you currently have GitHub Copilot, Codex, or any other GenAI tool enabled, you need to **disable it for completing this lab**.

You may use Generative AI to aid in **troubleshooting your code** without uploading the given source file. Do not upload the assignment directions and ask it to generate the solution or parts of the solution for you.
Uploading the assignment or given .cpp file to generate code, fix the given code, or enabling code generation/completion within VSCode will result in an academic integrity violation charge.

What you Can't DO/Use in this lab

If you do any of the following, you will receive a 0 on some or all of the assignment and possibly get charged with an academic integrity violation (this is very serious).

1. You may not use more advanced code or syntax in your lab assignment **that has not been taught** in CSC 1310 class **without referencing your source**. How do you reference your source? By providing a link/URL in a multi-line comment in the section of code where you use the advanced code. **Otherwise, this will be considered plagiarism.** You should be able to explain every line of code that is in your submission.
2. Look at syllabus to review what you may not (ever) use the following in your programming & lab assignments in CSC 1310

3. You **may not use generative AI (genAI)** to generate code, but you can use it to help in troubleshooting problems as described in the Generative AI Policy above.
4. You may not share your code with anyone.

DON'T FORGET TO FILL OUT THE LAB REPORT

After you have completed this assignment, fill out the lab report and submit your lab report in your zip file along with your source files. Name your screenshot **lab6ReportProof**.

Lab Report Link: https://tntech.co1.qualtrics.com/jfe/form/SV_0iJcCTCBgguPDpQ

WHAT TO TURN IN TO CANVAS

Create a **zip** file named **username_lab6.zip** containing the following files and upload it the **Lab 6** assignment. Replace username with your TTU. Example: `acrockett43_lab6.zip`

- **Driver.cpp**
- **LinkedList.h**
- **Track.h**
- **Dataset.csv**
- **Lab6ReportProof**

SPECIFICATIONS

GIVEN FILES

- **Driver.cpp** – this file is mostly written for you. However, there are comments with
  //!!!!! LOOK !!!!!
  where you need to add content.
- **dataset.csv** – I downloaded this data set of Spotify music tracks from https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset

SPOTIFY TRACK CLASS (TRACK.H)

- Design a Track class (**Track.h**) to hold a Spotify track.
- You should have a minimum of 4 attributes. Two must be **track_id** and **popularity**. Then, you must **choose 2 or more** other pieces of data from the data set (dataset.csv) to store as attributes. When I created my **Track** class, I chose track_id, artist, album_name, track_name, popularity (int), explicit (bool), and track_genre.
- You should have the following member functions:
  - **Default constructor** (can be empty but is required or your code won't compile when creating a new node in the appendNode function in the LinkedList class)
  - **Overloaded constructor**
  - **Overloaded << operator** friend function

LINKED LIST CLASS (LINKEDLIST.H)

- Design your own linked list class (**LinkedList.h**) to hold a sequence of Tracks.
- Implement the LinkedList class in a single header file and create all member functions inline.
- The linked **list node** should be implemented as a **struct** and the value inside the struct should be a Track object.
- The class should have a **constructor** and **appendNode()** function.
- You should also have a **displayNode()** function where it accepts an integer position number and it will display the Track object at that particular position in the LinkedList.
  - The head of the linked list is position 0.
- Don't forget to add a **destructor** that destroys the list.

DRIVER.CPP

- Read through the driver (**Driver.cpp**) to see what is going on. Make sure you understand the code. I put lots of comments. It would be good to undertstand how to read from a .csv file and parse through it. There are always weird things that you have to deal with in .csv files or really any data files.
- Complete the Driver by adding in code where the **!!!!! LOOK !!!!!** comments are throughout the file.
- Compile your code with **-Wall** to ensure you have no warnings or errors.

SAMPLE OUTPUT

User input is highlighted in **yellow**.

```
Reading from the dataset.csv file and adding Spotify Tracks to Linked List

The list now has 114000 nodes with Spotify Music Tracks.

Which Spotify Music Track would you like to display?
Enter 0 through 113999:  -4

Invalid Input. Enter 0 through 113999:  115000

Invalid Input. Enter 0 through 113999:  blah

Invalid Input. Enter 0 through 113999:  0

The Spotify Music Track at position 0 is:
    Track ID:  5SuOikwiRyPMVoIQDJUgSV
      Artist:  Gen Hoshino
       Album:  Comedy
  Track Name:  Comedy
  Popularity:  73
     Explicit: no
        Genre: acoustic


Display another? Enter y or n:  blah

Invalid input. Enter y or n:  q

Invalid input. Enter y or n:  y

Which Spotify Music Track would you like to display?
Enter 0 through 113999:  114000

There is no Spotify Music Track at that position.


Display another? Enter y or n:  y

Which Spotify Music Track would you like to display?
Enter 0 through 113999:  113999

The Spotify Music Track at position 113999 is:
    Track ID:  2hETkH7cOfqmz3LqZDHZf5
      Artist:  Ces├íria Evora
       Album:  Miss Perfumado
  Track Name:  Barbincor
  Popularity:  22
     Explicit: no
```

```
        Genre: world-music


Display another? Enter y or n:  3948

Invalid input. Enter y or n:  y

Which Spotify Music Track would you like to display?
Enter 0 through 113999:  3948

The Spotify Music Track at position 3948 is:
    Track ID:  6eGTANo4ak5Tb1FCBLtoFT
      Artist:  Los Rodriguez
       Album:  M├╢sica para casa
  Track Name:  Mi Enfermedad
  Popularity:  0
     Explicit: no
        Genre: alternative


Display another? Enter y or n:  y

Which Spotify Music Track would you like to display?
Enter 0 through 113999:  44286

The Spotify Music Track at position 44286 is:
    Track ID:  3pxCXv0NACYfYnLjFcQW5D
      Artist:  Saving Abel
       Album:  Saving Abel
  Track Name:  Addicted
  Popularity:  70
     Explicit: yes
        Genre: grunge


Display another? Enter y or n:  y

Which Spotify Music Track would you like to display?
Enter 0 through 113999:  10941

The Spotify Music Track at position 10941 is:
    Track ID:  5p2g5mraWmHLaxyofZ1UG6
      Artist:  Dub Pistols;Ed Solo
       Album:  Dub Pistols present Welcome To The Jungle (DJ Mix)
  Track Name:  Camberwell Carrot - Ed Solo Remix - Mixed
  Popularity:  11
     Explicit: no
        Genre: breakbeat


Display another? Enter y or n:  n
```